



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
SEMESTER I - 2020/2021

MAT1073: Mathematics for Operations Research
End Semester Examination



Time allowed: THREE Hours

Answer All Questions

1. (a) Show that the set

$$S = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{pmatrix} \mid x_1, x_2 \text{ are real numbers} \right\}$$

is a vector space over $\mathbb{R}$ with respect to the standard addition and scalar multiplication in $\mathbb{R}^3$:

Addition:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ y_1 + y_2 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z_1 + z_2 \end{pmatrix}$$

where $z_1 = x_1 + y_1$ and $z_2 = x_2 + y_2$.

Multiplication:

$$r \begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{pmatrix} = \begin{pmatrix} rx_1 \\ rx_2 \\ rx_1 + rx_2 \end{pmatrix}; \quad \text{where } r \in \mathbb{R}.$$

- (b) Determine whether the following sets form a subspace of $\mathbb{R}^3$.

- i. $S_1 = \{(x_1, x_2, x_3) \mid x_1 - x_2 + x_3 = 0\}$,
- ii. $S_2 = \{(x_1, x_2, x_3) \mid x_1 + 3x_2 + x_3 = 1\}$.

- (c) Prove that the set $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

(100 marks)

2. (a) Let $M = \begin{pmatrix} -1 & 3 \\ 4 & -2 \end{pmatrix}$, $u = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $v = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$.

Determine whether u and v are eigenvectors of M .

- (b) Find the eigenvalues and the corresponding eigenvectors of the matrix A ;

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ -2 & 2 & 1 \end{pmatrix}$$

Hence find the eigenvalues of A^{-1} .

(100 marks)

3. (a) In a market survey three commodities A, B, and C were considered. In finding out the total consumption number, some fixed weights were assigned to the three varieties of the commodities. The following table provides the information regarding the consumption of three commodities according to the three varieties and also the total weight received by the commodity.

Commodity	Variety			Total Weight
	I	II	III	
A	3	2	4	28
B	1	1	2	13
C	2	2	1	14

Formulate a mathematical model to find the weights assigned to the three varieties. Solve the formulated model by using Cramer's Rule.

- (b) i. Consider the following matrix A :

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 4 & 3 \end{pmatrix}$$

Find the inverse of matrix A using the elementary row operations.

- ii. Express the following system of linear equations in the form $A\underline{x} = B$ where A , $\underline{x}$ and B are matrices in appropriate dimensions.

$$x + y + 2z = 2$$

$$x + 2y + 2z = 1$$

$$2x + 4y + 3z = 3$$

- iii. Using the inverse of matrix A found in part (i), solve the above system of linear equations.

(100 marks)

4. (a) Solve the following system of linear equations using the Gaussian elimination method.

$$x_1 + x_2 - 2x_3 = 4$$

$$x_1 + 3x_2 - x_3 = 7$$

$$2x_1 + x_2 - 5x_3 = 7$$

- (b) Let

$$B = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -2 & 3 \\ -2 & -4 & 6 \end{pmatrix} \quad \text{and} \quad \underline{b} = \begin{pmatrix} 1 \\ 6 \\ 12 \end{pmatrix}$$

Use LU-decomposition of B to solve $B\underline{x} = \underline{b}$.

(50 marks)

5. (a) Solve the following homogeneous differential equation;

$$y^2 dx + (xy + x^2) dy = 0$$

- (b) Show that the following differential equation is exact, and hence solve it.

$$(2y \sin x + 2) dy + (y^2 \cos x - \sin x) dx = 0$$

- (c) Solve the following differential equation by reducing the equation into linear form.

$$xy - \frac{dy}{dx} = y^3 e^{-x^2}$$

(50 marks)

.END.